

Study Plan for PyQt Research Documentation

Date	Time	Summary	Description
2026-01-12 (Mon)	10:00 AM - 12:00 PM	Study PyQt: Setup & Basics	Initial setup of PyQt6 environment. Review PyQt5 vs PyQt6 differences. Explore main modules: QtWidgets, QtGui, QtCore.
2026-01-19 (Mon)	10:00 AM - 12:00 PM	Study QtWidgets: QMainWindow	Study the QMainWindow class. Learn about creating subclasses and using setCentralWidget.
2026-01-26 (Mon)	10:00 AM - 12:00 PM	Study QtWidgets: QApplication	Study the QApplication class. Learn about CLI enablement using sys.argv and starting the event loop with app.exec().
2026-02-02 (Mon)	10:00 AM - 12:00 PM	Study QtWidgets: QWidget	Study the QWidget class. Learn about creating windows and controlling their visibility. Understand setFixedSize.
2026-02-09 (Mon)	10:00 AM - 12:00 PM	Study QtWidgets: QPushButton	Study the QPushButton class. Understand how to create and customize buttons.
2026-02-16 (Mon)	10:00 AM - 12:00 PM	Study: Layouts - QSplitter	Study QSplitter layout. Learn about creating resizable sections for widgets.
2026-02-23 (Mon)	10:00 AM - 12:00 PM	Study: Layouts - QDockWidget	Study QDockWidget layout. Learn about creating dockable panels.
2026-03-02 (Mon)	10:00 AM - 12:00 PM	Study: Layouts - QVBoxLayout	Study QVBoxLayout. Implement thumbnails in a vertical list.
2026-03-09 (Mon)	10:00 AM - 12:00 PM	Study: Layouts - QGridLayout	Study QGridLayout. Implement thumbnails in a grid layout.
2026-03-16 (Mon)	10:00 AM - 12:00 PM	Study: GStreamer Integration	Study GStreamer pipeline rendering in PyQt. Focus on VideoOverlay and QWidget.

2026-03-23 (Mon)	10:00 AM - 12:00 PM	Study: GStreamer Pipelines	Study GStreamer pipeline options: Software Decode, Linux X11 sink, and decodebin.
2026-03-30 (Mon)	10:00 AM - 12:00 PM	Study: Appsink & QImage	Study the appsink -> QImage -> QLabel/QOpenGLWidget approach for GStreamer integration.
2026-04-06 (Mon)	10:00 AM - 12:00 PM	Study: Qt Feed Title Aspect	Study recreating the current feed title aspect in Qt using QWidget, QLabel, and QLayout.
2026-04-13 (Mon)	10:00 AM - 12:00 PM	Study: Dynamic UI - Layouts	Study creating a more immersive/dynamic look with QSplitter and QDockWidget.
2026-04-20 (Mon)	10:00 AM - 12:00 PM	Study: Dynamic UI - State & Style	Study state-driven UI with signals/slots, and styling with Qt Stylesheets and QPropertyAnimation.
2026-04-27 (Mon)	10:00 AM - 12:00 PM	Study: Non-Blocking Tasks	Study using QThread, QRunnable/QThreadPool, and QTimer for non-blocking tasks.